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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Job Had Four Bros
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG 8, SDG 9, SDG 12, SDG 16
Project / Solution Name	CoSolve
Name (Leader)	Wong Ye Yang
Email & Phone (Leader)	yeyangwong@gmail.com | 0103714818
University Affiliation(s)	Sunway College
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Customer-support teams often receive complaints through several channels and must read long messages, identify urgency, decide a fair next step, and draft a response quickly. This work is repetitive and inconsistent: an urgent shipping-damage complaint may be delayed, while staff may apply different categories or compensation guidance to similar cases. Customers experience slow, uneven resolution, while businesses face higher support costs, missed escalation signals, and limited visibility into recurring product or delivery issues. CoSolve focuses on complaint handling as a practical workflow problem: give staff one place to receive, assess, document, and resolve cases with a clear audit trail.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

CoSolve is a complaint-resolution workspace for customer-support teams. Staff can capture a complaint, review order and customer context, classify the issue, receive resolution guidance, draft a customer response, and record the resulting actions. The application combines a fine-tuned DistilBERT sentiment model with structured AI assistance for categorisation and response drafting. The sentiment model identifies positive, neutral, or negative customer language and returns a confidence score so uncertain cases can be routed for human review. The planned prototype uses a web interface, database-backed case records, API integrations, authenticated staff access, and cloud-hosted inference. AI output supports staff decisions; it does not automatically approve refunds or send customer messages.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: React and TypeScript. Backend: Node.js, Express, and TypeScript. Data layer: Drizzle ORM with the project database. Authentication: Clerk. AI: a fine-tuned DistilBERT sentiment classifier deployed as a Runpod Serverless endpoint, plus a Grafilab OpenAI-compatible language-model API for structured support assistance and response drafting. Deployment and development: Replit. The flow is: staff submits or imports a complaint -> API stores the case -> the API calls Runpod for sentiment -> staff reviews the case with AI guidance -> approved actions and customer communication are recorded in an audit trail. The solution is being built as a functional web prototype rather than a presentation-only concept. Official SDC sponsor-provided resources: to be confirmed by the team leader.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

The team is applying an iterative workflow: start from real complaint examples, map the support agent journey, identify pain points in triage and response consistency, then test the interface on representative cases. The prototype keeps a human in the loop for high-impact actions: staff review classifications, review recommended resolutions, and approve compensation or customer communication before it is recorded. Deployment is planned around separated frontend, backend, database, and model-inference components so the system can be tested end to end. Phase 2 feedback will be used to improve label quality, confidence thresholds, resolution policies, and the usability of the agent workspace.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 8.2: improve productivity through technological upgrading; SDG 9: support innovation through accessible digital infrastructure; SDG 12: encourage responsible operational practices; SDG 16.6: strengthen effective, accountable, and transparent processes. CoSolve supports these goals through more consistent, traceable customer-support operations.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Prototype baseline: complaint review is manual and case information can be fragmented across messages, order context, and staff notes. Baseline measurement will be established from a small set of representative cases during Phase 2.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Median time from complaint intake to staff-ready classification; percentage of cases with a recorded category, sentiment, confidence score, recommendation, and audit event.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Phase 2 target: reduce the median time needed to prepare a staff-reviewed complaint assessment by 30% on the prototype test set, while keeping human approval for consequential actions.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☒ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☒ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☒ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☒ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

Declared below.

Name	Type (code/dataset/model)	License
React, Node.js, Express, TypeScript, Drizzle ORM	code	Respective open-source licences
DistilBERT / Hugging Face Transformers	model / library	Apache-2.0
Womens Clothing E-Commerce Reviews dataset	dataset	Public dataset; attribution and licence to be confirmed
Restaurant reviews.csv	dataset	Project-provided review dataset; source/licence to be confirmed

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



23/10/2026